TALHA KAYAR

Address: Ankara, Türkiye **Phone:** +90 506 028 64 83 **Email:** talhakayar7@gmail.com

PROFILE SUMMARY

GRE Quantitative: 170/170 | Top 0.1% in National Exam Rankings.
Electrical Engineering senior specializing in Wireless Communications and Signal Processing. Combining elite quantitative aptitude with hands-on experience in SDR-based modulation classification and GNSS-free positioning. Skilled in bridging the gap between theoretical models and real-time hardware using Python, MATLAB, and C.

EDUCATION

Middle East Technical University (METU), Ankara *September 2021 - July 2026*
B.S. in Electrical and Electronics Engineering *(Expected)*
CGPA: 2.95/4.00, Top 23%

Key Courses:

- Telecommunications II (EE436) **In Progress**
- Introduction to Computer Networks (EE444) **In Progress**
- Telecommunications I (EE435) **AA**
- Signals and Systems I (EE301) **BA**
- Probability and Random Variables (EE230) **BA**
- Vector Space Methods in Signal Processing (EE499) **BB**

STANDARDIZED TESTS

GRE General Test *January 2026*
Quantitative: **170/170** | Verbal: 156/170 | Analytical Writing: 3.5/6.0

IELTS Academic *November 2025*
Overall: **8.0** (Reading: 9.0, Listening: 9.0, Speaking: 8.0, Writing: 6.0)

YKS (Higher Education Institutions Examination) (Turkey) *June 2021*
Ranked **1908**th out of **2.5 Million** Students **(Top 0.1%)**

ALES (Graduate Education Entrance Exam) (Turkey) *November 2025*
Ranked **1004**th in Quantitative **(Top 0.7%)**

EXPERIENCE

Roketsan July 2025 - August 2025
R&D Intern (GNSS & Positioning Systems) *Ankara, Türkiye*

- Developed a satellite selection algorithm using **Python** and **RTKLIB** to optimize positioning accuracy in NLOS scenarios.
- Mitigated positioning errors caused by atmospheric delays and vehicle dynamics (Yaw/Pitch/Roll).
- Analyzed GNSS signal integrity and implemented data fusion techniques for robust navigation.
- Built scalable preprocessing pipelines to handle large-scale satellite data logs.

INFINIA June 2024 - July 2024
Embedded Software Developer (Intern) *Ankara, Türkiye*

- Developed IoT communication solutions using **ESP32-S3** and **MQTT** protocol.
- Implemented real-time 2-way data transmission over Wi-Fi networks using Embedded C.

DataLobster March 2022 - April 2025
Signal Processing Undergraduate Researcher (Part Time) *Remote*

· Designed algorithms to **identify anomalies** in industrial sensor data streams.
· Applied **statistical signal processing methods** (Filtering, Spectral Analysis) for data analysis.
· Developed **Autoencoder models** and data pipelines for predictive maintenance solutions.

KEY ACADEMIC PROJECTS

SIRIUS: Outdoor Positioning System via Sensor Fusion (Capstone Project)

· Engineered an outdoor positioning system **without using GNSS** and integrating **Wi-Fi Fine Time Measurement (FTM)** and **IMU data**.
· Developed a Sensor Fusion algorithm using **Extended Kalman Filters** to combine inertial measurements with FTM ranging.
· Implemented a Machine Learning algorithm using **Random Forest Regression** to decrease the error rate from **10% to 2%** in positional estimations.
· Mitigated signal degradation caused by **NLOS (Non-Line-of-Sight) propagation** using **statistical error correction**.

SDR-Based Modulation Classification (Course Project)

· Developed a modulation classification system using **ADALM-PLUTO SDR** and MATLAB/Python.
· Implemented a **Decision Tree** algorithm based on statistical feature extraction (Variance, Kurtosis) to identify analog modulations (AM, FM, SSB, DSB).

Frequency Hopping Spread Spectrum Communication System (Course Project)

· Implemented a receiver and transmitter pair using **Frequency Hopping Spread Spectrum** in **MATLAB**.
· Applied **Spectral Analysis** and **Pilot-Aided Detection** to detect and analyze the signals.
· Used **Frequency-shift keying** to encode and decode various message signals throughout different channels.

CERTIFICATES & AWARDS

- **Fundamentals of Deep Learning** - Nvidia May 2024
- **Structuring Machine Learning Projects** - DeepLearning.AI January 2024
- **Neural Networks and Deep Learning** - DeepLearning.AI October 2023
- **Improving Deep Neural Networks: Hyperparameter Tuning, Regularization** - DeepLearning.AI October 2023
- **SQL for Data Science** - UC Davis February 2023
- **Getting Started with Data Analytics on AWS** - Amazon Web Services October 2022
- **TÜBİTAK Project Competition 2204-A** - 2nd Place in Mathematics 2018